Fluid Machinery Two marks

Unit – I

1. Define the term Impact of Jet.

Force is obtained from Newton's second law of motion or from impulse momentum equation. Thus impact of jet means the force exerted by the jet on a plate which may be stationary or moving.

i. Force exerted by the jet on a stationary plate when
   a. Plate is vertical to the jet
   b. Plate is inclined to the jet
   c. Plate is curved
ii. Force exerted by the jet on a moving plate when
   a. Plate is vertical
   b. Plate is inclined
   c. Plate is curved

2. Definition of a turbo machine

We classify as turbo machines all those devices in which energy is transferred either to, or from, a continuously flowing fluid by the *dynamic action* of one or more moving blade rows.

The word *turbo* or *turbinis* is of Latin origin and implies that which spins or whirls around. Essentially, a rotating blade row, a *rotor* or an *impeller* changes the stagnation enthalpy of the fluid moving through it by either doing positive or negative work, depending upon the effect required of the machine.

3. FORCE IS EXERTED BY A JET ON A STATIONARY VERTICAL PLATE

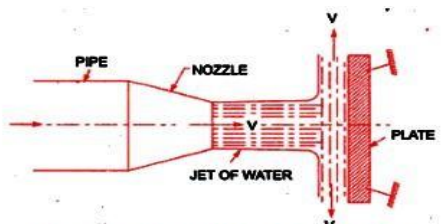

Fig. 17.1 Force exerted by jet on vertical plate.

$F_x = \rho a V^2$

$F_y = 0$

4. FORCE IS EXERTED BY A JET ON A STATIONARY INCLINED FLAT PLATE

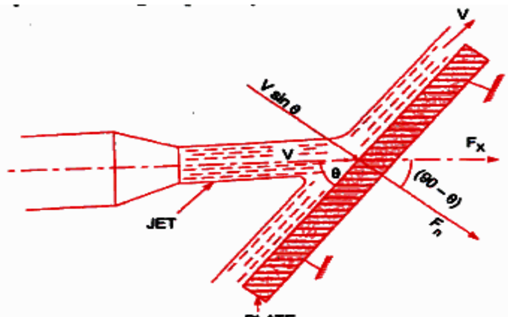

Fig. 17.2 Jet striking stationary inclined plate.

$F_x = \rho a V^2 \sin^2 \theta$

$F_y = \rho a V^2 \sin \theta \cos \theta$

5. **FORCE IS EXERTED BY A JET ON A STATIONARY CURVED PLATE At the Centre**

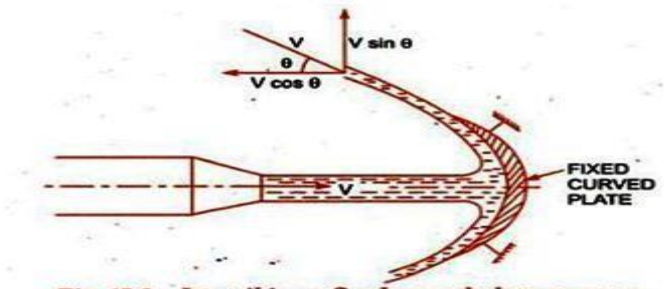

Fig. 17.3 *Jet striking a fixed curved plate at centre.*

Fx = $\rho a V^2 [1 + \cos\theta]$

Fy = $-\rho a V^2 \sin^2\theta$

6. **FORCE EXERTED BY A JET ON A HINGED PLATE**

X = distance of the centre of jet from hinge O, $\theta = angle\ of\ swing\ about\ hinge$

W = weight of plate acting at centre of gravity of the plate.

1. Force due to jet of water, normal to the plate,

   Fn = $\rho a V^2 \cos\theta$

2. Weight of the plate, W

   W = $\frac{aV^2}{\sin\theta}$

7. **Define Efficiency Of the jet.**

It is defined as the ratio of work done per second on the vane by initial kinetic energy per second of the jet.

Efficiency = $\frac{Work\ done\ per\ second}{Kinetic\ Energy\ per\ second}$

$= \frac{\rho aV\ r1\ [Vw\ 1 \pm Vw\ 2]\ X\ u}{\frac{1}{2}\rho a\ V3}$

8. **Efficiency of series vane**

It is defined as the ratio of work done per second on the vane by initial kinetic energy per second of the jet.

Efficiency = $\frac{Work\ done\ per\ second}{Kinetic\ Energy\ per\ second}$

$$= \frac{2[V-u]}{V^2}$$

Condition for maximum efficiency,

$$u = \frac{V}{2}$$

Maximum Efficiency = 0.5 = 50 %

Unit – II

1. Define Hydraulic Turbines.

Hydraulic turbines are the machines which use the energy of water and convert it to mechanical energy. The mechanical energy developed by a turbine is used in running an electric generator which is directly coupled to the shaft of the turbine. The electric generator thus develops electric power, which is known as hydro-electric power.

2. Give example for a low head, medium head and high head turbine.

Low head turbine – Kaplan turbine
Medium head turbine – Modern Francis turbine
High head turbine – Pelton wheel

3. What is impulse turbine? Give example.

In impulse turbine all the energy converted into kinetic energy. From these the turbine will develop high kinetic energy power. This turbine is called impulse turbine. Example: Pelton turbine

4. What is reaction turbine? Give example.

In a reaction turbine, the runner utilizes both potential and kinetic energies.
Here portion of potential energy is converted into kinetic energy before entering into the turbine.
Example: Francis and Kaplan turbine.

5. What is axial flow turbine?

In axial flow turbine water flows parallel to the axis of the turbine shaft.
Example: Kaplan turbine

6. What is mixed flow turbine?

In mixed flow water enters the blades radially and comes out axially, parallel to the turbine shaft.
Example: Modern Francis turbine.

7. What is the function of spear and nozzle?

The nozzle is used to convert whole hydraulic energy into kinetic energy. Thus the nozzle delivers high speed jet. To regulate the water flow through the nozzle and to obtain a good jet of water spear or nozzle is arranged.

8. Define gross head and net or effective head.

Gross Head: The gross head is the difference between the water level at the reservoir and the level at the tailstock.

Net Head: It is also called effective head and it is defined as head available at inlet of the turbine. It is difference between Gross Head to Head losses due to Friction.

9. Efficiency of the Turbines

The following are the important efficiencies of a turbine

a) Hydraulic Efficiency, η_h
b) Mechanical Efficiency, η_m
c) Volumetric Efficiency, η_v
d) Overall Efficiency, η_o

10. Define Hydraulic Efficiency of the Turbine.

It is the defined as ratio of power given by water to the runner of a turbine to the power supplied by the water at the inlet of the turbine.

$$\eta_h = \frac{\text{Power delivered to runner}}{\text{Power supplied at inlet}} = \frac{R.P}{W.P}$$

R.P = Q [$V_{w1} \pm V_{w2}$] x u kW ....for Pelton Turbine

R.P = Q [$V_{w1}\,u_1 \pm V_{w2}\,u_2$] kW ....for a radial flow turbine

W.P = g x Q x H kW ......when flowing fluid is water

W.P = $\frac{\rho g Q H}{1000}$ kW.. when flowing fluid is other than the water

11. Define Mechanical Efficiency of the Turbine.

It is defined as the ratio of Shaft power of the turbine to the Runner power

$$\eta_m = \frac{\text{Power at the shaft of the turbine}}{\text{Powe eliverd by the water to the runner}} = \frac{S.P}{R.P}$$

12. Define Volumetric Efficiency of the Turbine

It is defined as the ratio of volume of water actually striking the runner to the volume of water supplied to the turbine.

$$\eta_v = \frac{\text{Volume of water actually striking the runner}}{\text{Volume of water supplied to the turbine}}$$

13. Define Overall Efficiency of the Turbine.

It is defined as the ratio of power available at the shaft of the turbine to the power supplied by the water at inlet of the turbine.

$$\eta_o = \frac{\text{Shaft Power}}{\text{Water Power}} = \frac{S.P}{W.P}$$

$$\eta_o = \eta_m \times \eta_h$$

14. What are the general classifications of Turbines.

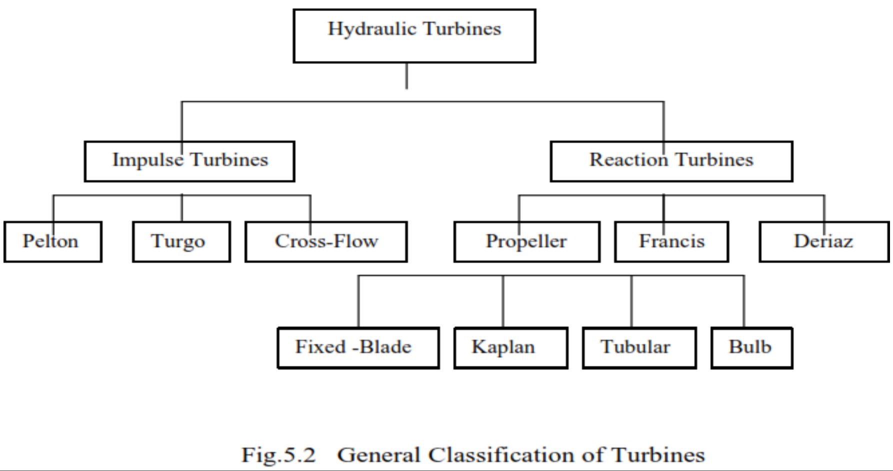

Fig.5.2 General Classification of Turbines

15. What are the advantages and disadvantages of multi jet Pelton Wheel

Advantages of multi-jet:

- Higher rotational speed
- Smaller runner
- Less chance of blockage

Disadvantages of multi-jet:

- Possibility of jet interference on incorrectly designed systems
- Complexity of manifolds

16. List the Parts of the Pelton Wheel.

Pelton wheel consists of the following main parts

1. Penstock.
2. Spear and nozzle.
3. Runner with buckets.
4. Break nozzle.
5. Outer casing.
6. Governing mechanism.

17. What is a Breaking Jet?

When the Nozzle is completely closed by moving the spear in the forward direction, the amount of water striking the runner reduces to zero. But the runner due to inertia goes on revolving for a long time. To stop the runner in a short time, a small nozzle is provided which directs the jet of water on the back of the vane. This jet of water is called Breaking Jet.

18. Define Jet Ratio.

It is defined as the ratio of pitch Diameter (D) of the Pelton Wheel to the Diameter of the jet (d). It is denoted by 'm' and is given by

m = $\frac{}{d}$ (= 12 for Most Cases)

19. Define: Speed ratio.

It is the ratio of peripheral speed at outlet (u_2) to the theoretical velocity of jet corresponding to manometric head H_m.

$$Ku = \frac{u2}{\sqrt{2gHm}}$$

Ku var $iesfrom 0.95 to 1.25$.

20. Define: Flow ratio.

It is the ratio of the velocity of flow at exit (V_{f2}) to the theoretical velocity of the jet corresponding to manometric head (H_m).

$$Kf = \frac{Vf\,2}{\sqrt{2gHm}}$$

Ku var $iesfrom 0.1 to 0.25$.

21. What is the function of governing mechanism in pelton wheel?

Governing mechanism is used to regulate the water flow to the turbine at constant level so that the speed of the turbine kept constant. This automatically regulates the quantity of water flowing through the runner in accordance with any variation of load.

22. What is draft tube and explain its function?

After passing through the runner, the water is discharged to the tailrace through a gradually expanding tube called draft tube.

The pressure at the exit of the runner of a reaction turbine is generally less than atmospheric pressure. By passing reduced through draft tube, the outer velocity of water is reduced and gain in useful pressure head is achieved to increase the output of turbine.

23. What are the significant of unit quantities and specific quantities?

1. To predict the behaviour of a turbine working under different conditions.

2. Make comparison between the performances of turbine of same types of different sizes.

24. Define the specific speed of a turbine.

The specific speed of any turbine is the speed in r.p.m of a turbine geometrically similar to the actual turbine but of such a size that under corresponding conditions it will develop 1 metric horsepower when working under unit head.

$N_s = N\sqrt{P} / H^{5/4}$

Where Ns = specific speed

P = power in HP

25. Differentiate between the turbines and pumps.

Sl. No.	Turbines	Pumps
1.	Turbine converts hydraulic energy into mechanical energy.	Pump converts mechanical energy into hydraulic energy.
2.	It is energy producing machine.	It is energy absorbing machine.
3.	Flow takes place from high-pressure side to the low-pressure side.	Flow takes place from low-pressure side to the high-pressure side.
4.	Flow is decelerated.	Flow is accelerated.

26. Sketch the Types Of Draft Tubes.

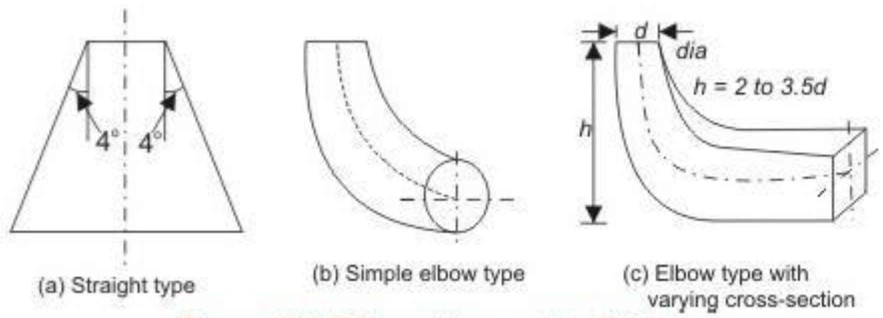

Figure 30.4 Different types of draft tubes

27. Define Degree of Reaction.

Degree of Reaction is defined as the ratio of pressure energy change inside a runner to the total energy change inside the runner. It is represented by ’R’. Hence mathematically it is written as

$$R = \frac{Chang\ f\ pressure\ \ energy\ \ inside\ \ the\ Turbine}{Change\ \ of\ \ total\ \ energy\ \ inside\ \ the\ turbine}$$

$$R = 1 - \frac{(V_1^2 - V_2^2)}{2g\,H_e}$$

28. Degree of reaction for pelton wheel.

For a pelton wheel turbine,

$u_1 = u_2$ and $V_{r1} = V_{r2}$

$$R = 1 - \frac{(V_1^2 - V_2^2)}{(V_1^2 - V_2^2)} = 0$$

29. Degree of Reaction for actual Reaction turbine

$$R = 1 - \frac{\cot\alpha}{2(\cot\alpha - \cot\theta)}.$$

30. Define Efficiency of the Draft Tube.

The efficiency of the draft tube is defined as the ratio of actual conversion of kinetic head in the draft tube to the kinetic head at the inlet of the draft tube. Mathematically, it is written as

$$\eta_d = \frac{Actual\ \ conversion\ \ of\ kinetic\ \ head\ into\ the\ pressure\ head}{Kinetic\ \ head\ \ at\ the\ inlet\ \ of\ the\ draft\ \ tube}$$

31. What are the Uses of the draft tube?

- It permits a negative head to be established at outlet of the runner and thereby increase the net head on the turbine.
- It converts a large proportion of the kinetic energy rejected at the outlet of the turbine into useful pressure energy.
- The net head on the turbine increases.
- Develops more power and also efficiency of the turbine increases.

32. Define Unit Speed.

It is defined as the speed of the turbine working under unit head. It is denoted by ‘N_u’. The expression of the unit speed is obtained as:

$$N_u = N/\sqrt{H}$$

33. Define Unit Discharge.

It is defined as the discharge passing through a turbine, which is working under a unit head. It is denoted by the symbol ‘Q_u’. The expression of the unit discharge is given by:

$$Q_u = Q/\sqrt{H}$$

34. Define Unit Power.

It is defined as the power developed by the turbine, working under a unit head. It is denoted by a symbol ‘P_u’. The expression for unit power is given by:

$$P_u = P/H^{3/2}$$

Unit – III

1. **What is meant by Hydraulic Pump?**

- A pump is device which converts mechanical energy into hydraulic energy.
- It is energy absorbing machine.
- Flow takes place from low-pressure side to the high-pressure side.
- Flow is accelerated.

2. List out the various components of a centrifugal pump.

a) Impeller

b) Casing

c) Suction pipe, strainer and foot valve

d) Delivery pipe and delivery valve

3. What are the types of casing for centrifugal pump?

i) Volute casing.

ii) Vortex casing.

iii) Volute casing with guide blades.

4. What is meant by priming of pumps?

The delivery valve is closed and the suction pipe, casing and portion of the delivery pipe up to delivery valve are completely filled with the liquid so that no air pocket is left. This is called as priming.

5. What do you mean by manometric efficiency and mechanical efficiency of a centrifugal pump?

It is the ratio between manometric head and head imparts by impeller to liquid.

$$\eta mano = \frac{Manometrichead}{Hea\dim partesbyimpellertoli\ quid}$$
$$= \frac{Outputofthepump}{Powerimpartedbytheim\ peller}$$

6. What is meant by cavitations?

It is defined as the phenomenon of formation of vapour bubbles of a flowing liquid in a region where the pressure of the liquid falls below its vapour pressure and the sudden collapsing of these vapour bubbles in a region of higher pressure.

7. How can we identify the cavitation in pumps?

i) Sudden drop in effiency.
ii) Head falls suddenly.
iii) More power requirement.
iv) Noise and vibrations.

8. State any two precautions against cavitations.

- The pressure should not be allowed to fall below its vapour pressure.
- Special material coatings can be given to the surfaces where the cavition occurs.

9. What is the principle of reciprocating pumps? And state its displacement type

It operates on a principle of actual displacement of liquid by a piston or plunger, which reciprocates in a closely fitting cylinder.

10. State the main classification of reciprocating pump.

- According to the liquid being in contact with piston or plunger.
- According to the number of cylinders provided.

11. Mention the main components of reciprocating pump.

- Piston or plunger.
- Suction and delivery pipes.
- Crank and connecting rod.

12. What is the main difference between single acting and double acting reciprocating pump?

In a single acting reciprocating pump, the liquids acts on one side of the piston only whereas in double acting reciprocating pump, the liquid acts on both sides of the piston.

13. Define "Slip" of reciprocating pump. When does the negative slip occur?

The difference between the theoretical discharge and actual discharge is called slip of the pump. Slip = Q_{th} - Q_{act}

$$Percentage slip = \frac{Qth - Qact}{Qth} \times 100$$

$$\left(1 - \frac{Qact}{Qth}\right) \times 100 = (1 - Cd) \times 100$$

But in sometimes, Q_{act} may be higher than Q_{th}, in such case C_d is greater than unity and the slip will be negative called as negative slip.

14. What are rotary pumps? Give examples.

Rotary pumps resemble like a centrifugal pumps in appearance. But, the working method differs. Uniform discharge and positive displacement can be obtained by using these rotary pumps. So, we can clearly say, it has the combined advantages of both centrifugal and reciprocating pumps. The various types of rotary pumps are

1. External pump.
2. Internal gear pump.
3. Lobe pump.
4. Vane pump.

15. What is the minimum starting speed of the centrifugal pump?

If the pressure raise in the impeller is more than or equal to manometric head, the centrifugal pump will start delivering the water.

$$N = \frac{120 \times \eta_{man} \times V_{w2} \times D_2}{[D_2^2 - D_1^2]}$$

16. Define specific speed of centrifugal pump.

The specific speed of the centrifugal pump is defined as the speed of a geometrically similar pump which would deliver one cubic metre of liquid per second against a head of one metre. It is denoted by 'N_s'.

$$N_s = N\sqrt{Q} / H_m^{3/4}$$

17. Define Characteristics curves of centrifugal pump.

Characteristic curves of centrifugal pumps are defined as those curves which are plotted from the results of a number of tests on the centrifugal pump.

- Main characteristics Curves
- Operating characteristics Curves.
- Constant Efficiency or Muschel Curves

18. What is iso-efficiency curve?

Plotting head versus discharge and efficiency versus discharge curves for different speed. The points having same efficiency are then joined by smooth curves. These smooth curves represent the iso- efficiency curves.

19. What is Thoma's Cavitation factor?

The Thoma's cavitation factor is given by

$$\varsigma = \frac{(H_b) - H_s - h_{LS}}{H}$$

20. Explain the classification of the reciprocating pump?

a. Single acting pump b. Double acting Pump

a. Single cylinder Pump b. Double cylinder Pump c. Triple Cylinder Pump

21. Define Indicator diagram.

The Indicator diagram for a reciprocating pump is defined as the graph between the pressure head in the cylinder and distance travelled by the piston from inner dead centre for one complete revolution of the crank.

23. Define and give the uses of air vessels.

Air vessel is a closed chamber containing compressed air in the top portion and liquid at bottom portion of the chamber.

i. To obtain a continuous supply of liquid at a uniform rate

ii. To save a considerable amount of work in overcoming the frictional resistance in the suction and delivery pipes.

24. What is the equation of Maximum suction lift?

$$H_s = H_a - H_v - \frac{V_s^2}{2g} - h_{fs}$$

25. Define Net Positive Suction Head (NPSH).

- The term NPSH is very commonly used in pump industry. Actually the minimum suction conditions are more frequently used in terms of NPSH.
- It is defined as the absolute pressure head at the inlet to the pump, minus the vapour pressure head plus the velocity head.

1. **Define Air Compressor.**

   Air compressor is a machine which compresses the air and raises its pressure. It sucks the air from the atmosphere, compresses it and then delivers the same under a high pressure.

2. **Classification of air compressor.**

   i. According to the working
      a. Reciprocating compressor b. Rotary compressor
   ii. According to action
      a. Single acting compressor b. Double acting compressor
   iii. According to number of stages
      a. Single stage compressors b. Multistage Compressors

3. **What is meant by single acting compressor?**

In single acting compressor, the suction, compression and delivery of air take place on one side of the piton.

4. **What is meant by double acting compressor?**

In double acting reciprocating compressor, the suction, compression and delivery of air take place on both side of the piston.

5. **What is meant by single stage compressor?**

   In single stage compressor, the compression of air from the initial pressure to the final pressure is carried out in one cylinder only.

6. **What is meant by multistage compressor?**

   In multistage compressor, the compression of the air from the initial pressure to the final pressure is carried out in more than one cylinder.

7. **What the advantages of multi stage compression are with inter cooling over single stage compression for the same pressure ratio?**

   1. The work done per kg of air is reduced in multistage compression with inter cooler as compared to single stage compression for the same delivery pressure.
   2. It improves the volumetric efficiency for the given pressure ratio.
   3. The size of the cylinders (i.e., high pressure and low pressure) may be adjusted to suit the volume and the pressure of the air.
   4. It reduces the leakage loss considerably.
   5. It gives more uniform torque and hence a smaller size flywheel is required.
   6. It provides effective lubrication because of lower operating temperature.
   7. It reduces the cost of the compressor.

8. **Define volumetric efficiency,**

   Volumetric efficiency is defined as the ratio of volume of free air sucked into the compressor per cycle to the stroke volume of the cylinder.

$$\eta_{Iso} = \frac{\text{Isothermal work}}{v_s}$$

v_a = suction volume

v_S = stroke volume

9. Define clearance ratio

Clearance ratio is defined as the ratio of clearance volume to swept volume (or) stroke volume

$$C = \frac{vc}{vs}$$

v_c = clearance volume

v_S = swept volume

10. Define isothermal Efficiency of air compressor.

It is defined as the ratio of isothermal work to Indicated work

Isothermal efficiency (Compressor efficiency)

$$\eta_{\text{Iso}} = \frac{Isothermalwork}{Indicatedwork}$$

11. Define isentropic efficiency

It is the ratio of the isentropic power to the brake power required to drive the compressor

$$\text{Isentropic efficiency} = \frac{Isentropicpower}{Actualbrakepower}$$

12. Define mean effective pressure. How it is related to indicated power of on IC engine.

Mean effective pressure is defined as hypothetical pressure, which is considered to be acting on the piston IP throughout the power stroke.

Indicated power, IP = P_m x L x A x N x n

Where P_m = Mean effective pressure kPa

A = Area m^2

N = rpm [N/2 for 4 stroke]

n = no. of cylinders

13. Explain how flow of air is controlled in a reciprocating compressor?

The flow of air is controlled by centrifugal governor, or by maintaining the speed of motor constant or by providing the air pocket advancement to the cylinder.

14. Mention the important applications of compressed air.

Compressed air used in

1. Pneumatic brakes
2. Pneumatic drills
3. Spray painting
4. Pneumatic Jacks
5. Air conditioning
6. Refrigeration

15. What factors limit the delivery pressure in a reciprocating compressor?

1. To obtain high delivery pressure, the size of the cylinder will be large.
2. Temperature of air.

16. Why clearance is necessary and what is its effect on the performance of reciprocating compressor.

When the piston reaches top dead centre in the cylinder, there is a dead space between piston top and cylinder head. This space is known as clearance space and the volume occupied by this space is known as clearance volume.

17. What is compression ratio?

Compression ratio is defined as the ratio between total volume and clearance volume.

$$\text{Compression ratio} = \frac{Totalvolume}{Clearancevolume}$$

18. What is meant by inter cooler?

An inter cooler is a simple heat exchanger. It exchanges the heat of compressed air from the low-pressure compressor to the circulating water before the air enters to the high-pressure compressor. The purpose of inter cooling is to minimize the work of compression.

19. Give the expression for work done for a multistage compressor with perfect inter cooling.

$$W = \frac{2n}{n-1}\, p1v1 \left[\left(\frac{(px+1)}{(p)}\right)^{\frac{n-1}{xn}} - 1\right] KJ$$

Where

χ ➔ no. Of stages
P_1➔ Initial pressure
v_1➔ Initial volume
n ➔ Index

20. Give the expression for work done for a two-stage compressor with perfect inter cooling.

$$W = \frac{2n}{n-1}\, p1v1 \left[\left(\frac{(p3)}{(p1)}\right)^{\frac{n-1}{2n}} - 1\right] KJ$$

21. Discuss the effect of clearance upon the performance of an air compressor?

The volumetric efficiency of air compressor increases with decreasing the clearance of the compressor.

The free air delivered by the compressor is increased by decreasing the clearance volume.

22. Give two merits of rotary compressor over reciprocating compressor

1. Rotary compressor gives uniform delivery of air compare to reciprocating compressor.
2. Rotary compressors are small in size for the same discharge compared with reciprocating compressors.
3. Lubricating system is more complicated in reciprocating compressor where as it is very simple in rotary compressor.

23. Explain the working principle of rotary compressor.

In rotary compressor the air is entrapped between two sets of engaging surfaces and the pressure rise is either by back flow of air (Roots blower) or by both squeezing action and backflow of air (vane type).

24. What are the types of rotary air compressor?

i. Roots blower compressor
ii. Vane blower compressor
iii. Centrifugal blower compressor
iv. Axial flow compressor

25. Define static and total head Quantities.

The total kinetic energy is converted into heat energy, which will increase the pressure and temperature. The new pressure and temperature of the air are called total heat or stagnation temperature and pressure respectively

26. What do you understand by the term slip factor?

The difference between blade velocity and whirl velocity is known as slip and the ratio of whirl velocity and blade velocity is known as slip factor.

27. What are the types of fans?

i. Centrifugal fans
ii. Drum type fans
iii. Partial flow fans

28. What are the losses in the fans?

i. Impeller entry losses
ii. Leakage losses
iii. Impeller losses
iv. Diffuser and volute losses
v. Disc friction

Unit – V

1. **Define fluid system.**

   Fluid system is defined as the device in which power is transmitted with the help of a fluid which may be liquid or gas under pressure. Most of these devices are based on the principles of fluid statics and fluid kinematics.

2. **What are the devices of the fluid system?**

   i. The hydraulic press
   ii. The hydraulic accumulator
   iii. The hydraulic intensifier
   iv. The hydraulic ram
   v. The hydraulic lift
   vi. The hydraulic crane
   vii. The fluid or hydraulic coupling
   viii. The fluid or hydraulic torque converter
   ix. The air lift pump
   x. The gear-wheel pump

3. **Define hydraulic press.**

   The hydraulic press is a device used for lifting heavy weights by the application of a much smaller force. It is based on Pascal's law, which states that the intensity of pressure in a static fluid is transmitted equally in all direction.

4. **Define mechanical advantage of hydraulic press.**

   The ratio of weight lifted to the force applied on the plunger is defined as the mechanical advantage. Mathematically mechanical advantage is written as

   M.A = W/F

5. **Define the hydraulic accumulator.**

   The hydraulic accumulator is a device used for storing energy of a liquid in the form of pressure energy, which may be supplied for any sudden or intermittent requirement.

6. **What is the capacity of accumulator?**

   It is defined as the maximum amount of hydraulic energy stored in the accumulator. The expression of the capacity of the accumulator is given by:

   Capacity of accumulator = p x A x L

7. **Define the Hydraulic Intensifier.**

   The device which is used to increase the intensity of pressure of water by means of hydraulic energy available from large amount of water at low pressure is called hydraulic intensifier.

8. **Define hydraulic ram and efficiency of the hydraulic ram.**

   The hydraulic ram is a pump which raises water without any external power of its operation. When large quantity of water is available at a small height, a small quantity of water can be raised to a greater height with the help of hydraulic ram.

   Efficiency $= \frac{Energy\ \ deliverd\ \ b\ \boxed{?}e\ ram}{Energy\ \ supplied\ \ by\ t\boxed{?}e\ ram} = \frac{w\ \ x\ H}{W\ x\ \boxed{?}}$

9. **Define hydraulic lift.**

The hydraulic lift is a device used for carrying passenger or goods from one place to another place in multi-stored building. The hydraulic lifts are two types namely,

i. Direct acting hydraulic lift

ii. Suspended hydraulic lift

10. **Define hydraulic crane.**

Hydraulic crane is a device, used for raising or transferring heavy loads. Its widely used in workshops, ware house and dock sidings.

11. **Define Fluid or hydraulic Coupling.**

The fluid or hydraulic coupling is a device used for transmitting power from driving shaft to driven shaft with the help of fluid, there are no mechanical connection between the two shafts.

12. **Define Hydraulic Torque converter.**

The hydraulic torque converter is a device used for transmitting increased torque at the driven shaft. The torque transmitted at the driven shaft may be more or less than the driving shaft.

13. **What is meant by air lift pump?**

The air lift pump is a device which is used for lifting water from a well or sump by using compressed air. The compressed air is made to mix with water. The density of the mixture of air and water is reduced.

14. **What is gear wheel pump?**

The gear wheel pump is a rotary pump in which two gears mesh to provide the pump action. This type of pump is mostly used for cooling water and pressure oil to be supplied for lubrication.

15. **What is self priming pump?**

Self priming pump is designed to lift water from low level below the pump suction without having to fill the suction pipe with liquid. Classification of self primers is

- Clear water self primers
- Solid handling self primers

16. **Define screw Pump.**

- Screw pumps are a more complicated type of rotary pumps, featuring two or three screws with opposing thread —- that is, one screw turns clockwise, and the other counter clockwise.
- The turning of the screws, and consequently the shafts to which they are mounted, draws the fluid through the pump.
- As with other forms of rotary pumps, the clearance between moving parts and the pump's casing is minimal.

17. **Define vane pump.**

A **rotary vane pump** is a positive-displacement pump that consists of vanes mounted to a rotor that rotates inside of a cavity. In some cases these vanes can be variable length and/or tensioned to maintain contact with the walls as the pump rotates.

18. **Define diaphragm pump.**

A diaphragm pump is a positive displacement pump that uses a combination of the reciprocating action of a rubber, thermoplastic or Teflon diaphragm and suitable non-return check valves to pump a fluid. Sometimes this type of pump is also called a membrane pump.

19. **Define vacuum pump.**

A **vacuum pump** is a device that removes gas molecules from a sealed volume in order to leave behind a partial vacuum. The first vacuum pump was invented in 1650 by Otto von Guericke, and was preceded by the suction pump, which dates to antiquity.

20. **Define jet pump.**

A device in which a small jet of steam, air, water, or other fluid, in rapid motion, lifts or otherwise moves, by its impulse, a larger quantity of the fluid with which it mingles.